Tianhao Li

tianhao.li | tianhao.li2@duke.edu | LinkedIn | Google Scholar | Orcid | Github

Academia Research Experiences

University of Wisconsin, Madison **October 2024 – Present**

Research Intern (Part-time) @ Safe And secure Foundation mOdel systems Lab (SaFoLab) *Wisconsin, United States*

- **Duties**: Led a research team to advance the AgentFuzzer project, proposing a unified agent schema descriptor, AgentSchemaDescriptor. Developed the first systematic fuzz testing framework for LLM agents, addressing the challenge of improving generalizability in agent safety alignment testing. (To be submitted to ICML'25)
- **Mentor**: Prof. Dr. Chaowei Xiao

Zhejiang University **July 2024 – October 2024**

Visiting Researcher (Part-time) @ Knowledge Engine Lab *Zhejiang, China*

- **Duties**: Led a 15-member research team on the SciSafeEval project, proposing the first safety benchmark for LLMs in scientific tasks. Identified misuse risks in SOTA LLMs (including GPT-4o, LLaMa3.2, QWen2.5) across biology, chemistry, medicine, and physics, and proposed mitigation strategies using Chain-of-Thought (CoT) and few-shot learning techniques.
- **Impacts**: arXiv:2410.03769 (ICLR'25 Sub) | Huggingface:Tianhao0x01/SciSafeEval | Github:DavidLee528/SciSafeEval
- **Mentors**: Prof. Dr. Qiang Zhang, Prof. Dr. Huajun Chen

High-Tech Innovation Institute **July 2022 – July 2023**

Research Assistant @ VR & AR Technology Innovation Center *Beijing, China*

- **Duties**: Led an 8-member R&D team developing a gait-based Intelligence, Surveillance, and Reconnaissance (ISR) system. Implemented a gait recognition SaaS prototype system and proposed security features including authentication, tracking, and early warning for physical attacks. Directed technical architecture design, technical reports, and research paper writing.
- **Impacts**: Applied 100K+ CNY funding, published 7 SCI/EI papers, won 4 national professional contests, delivered 3 oral presentations and 1 poster at international conferences.
- **Mentors**: Prof. Dr. Lijun Wang

Industry Research Experiences

TOPSEC **March 2024 – September 2024**

Security Researcher @ Department of Innovative Technology Research *Beijing, China*

- **Duties**: **(i)** Contributed to the development of a pioneering prototype system for machine learning model security assessment, focusing on the design and implementation of a robust backend evaluation module. Successfully debugged and mitigated adversarial and poisoning attacks, enhancing the system's resilience and accuracy.; **(ii)** Contribute to a phishing email detection project, overseeing dataset collection and enhancement initiatives. Successfully applied Large Language Models (LLMs) and Supervised Fine Tuning (SFT) to develop a robust detection system.
- **Supervisor**: GM of Innovation Research Dr. Wei Wang

NSFOCUS **September 2023 – March 2024**

Security Researcher @ Dubhe Lab, Innovation Institute *Beijing, China*

- **Duties**: **(i)** Design core algorithms and develop a prototype for NSFOCUS LLM Security Assessment System (LSAS); **(ii)** Enhance the security & privacy performance of NSFGPT and other private LLMs through offensive testing, i.e. red teaming; **(iii)** Conduct literature reviews encompassing research papers and Gartner's consulting reports on trustworthy AI, enriching the threat intelligence database; **(iv)** Write white paper on secure LLM.
- **Impacts**: The release of LSAS, SecLLM NSFGPT, and white paper Enhancing Network Security with SecLLM.
- **Supervisor**: Principal Security Researcher Dr. Xingkai Wang, CIO & GM of Innovation Research Dr. Wenmao Liu

Selected Publications

- Tianhao Li, Jingyu Lu, Chuangxin Chu, Tianyu Zeng, Yujia Zheng, Mei Li, Haotian Huang, Bin Wu, Zuoxian Liu, Kai Ma, Xuejing Yuan, Xingkai Wang, Keyan Ding, Huajun Chen, Qiang Zhang*. SciSafeEval: A Comprehensive Benchmark for Safety Alignment of Large Language Models in Scientific Tasks. *International Conference on Learning Representations, 2025 (ICLR Submission)* [PDF]
- Tianhao Li, Weizhi Ma, Yujia Zheng, Xinchao Fan, Guangcan Yang, Ying Li, Lijun Wang, Zhengping Li*. A Survey on Gait Recognition Against Occlusion: Taxonomy, Dataset and Methodology. *PeerJ Computer Science, 2024 (SCI, JCR-Q1, Accepted)* [Poster]

- Weizhi Ma, Ying Li*, Tianhao Li, Haowei Yang, Zhengping Li, Lijun Wang, Junyu Xuan. SFSWTS: EEG Emotion Recognition Based on Spatial-frequency Shifted Windows Time Self-attention Neural Network. *Pattern Recognition, 2024 (SCI, JCR-Q1, Peer Review)*
- Weizhi Ma, Yujia Zheng, Tianhao Li, Zhengping Li, Ying Li, Lijun Wang*. A comprehensive review of deep learning in EEG-based emotion recognition: classifications, trends, and practical implications. *PeerJ Computer Science, 2024 (SCI, JCR-Q1)* [PDF]
- Tianhao Li, Yujia Zheng, Weizhi Ma, Guangshuo Wang, Zhengping Li, and Lijun Wang*. Trustworthy Metaverse: A Comprehensive Investigation into Security Risks and Privacy Issues in Artificial Intelligence-Extended Reality Systems. *SID Symposium Digest of Technical Papers, 2024 (ICDT)*
- Yujia Zheng, Tianhao Li, Weizhi Ma, Jiaxiang Zheng, Zhengping Li*, and Lijun Wang. Unveiling Privacy Challenges: Big Data-Driven Digital Twins in Smart City Applications. *SID Symposium Digest of Technical Papers, 2024 (ICDT)*
- Tianhao Li, Yujia Zheng, Haoan Zhang, Weizhi Ma, and Ying Li*. Research on Real-time Early Warning Mechanism of Aviation Safety Based on Finite State Machine Underlying in QAR Stream Data. *Proceedings of the 2023 7th International Conference on Big Data and Internet of Things, 2023 (BDIoT)* [PDF][Oral][Slide]

Open Source Activities

- NVIDIA/garak (Generative AI Red-teaming Tool Kit. Metasploit in the field of LLM Security. 2.8K+ Stars):

  (i) Robustness Enhancement: Python interpreter version bug fix#296; Replicate LLM generator exception error fix#401; Add exception handling logic to the Detector base class#566; (ii) New Feature: Multi-modal Jailbreaking Attack on LLaVA #587 (Make it the **world's first** tool with multimodal red teaming capabilities.); Prompt Architecture Enhancement for Better Multi-modal Red Teaming (#658); Introducing new probe SciSafeEval (#948); Network Proxy Feature For Generator (garak:proxy_dev) (v0.9.0.9, v0.9.0.11, v0.9.0.13)

Academic Services

- Reliability Engineering & System Safety, Elsevier *(JCR-Q1, IF=9.4, Top)* — Reviewer
- The 31st International Conference on Computational Linguistics (COLING 2025) — Reviewer
- MLCommon/AI Risk & Reliability (AIRR) Working Group — Member/Individual Contributor

Honors and Awards

- Aug.2024, Second Prize in 17th CISCN Security Project Contest [certificate] — *Top7.2%, National*
- Jul.2024, Top 8 in Galaxy Generative AI Safety Contest Attack Track — *Top9.0%, National*
- Feb.2024, S Prize in 2024 Mathematical Contest In Modeling (MCM) [paper] — *Top?%, National*
- Nov.2023, Second Prize in 11th Digital Media Technology and Creativity Contest [certificate] — *Top9.8%, National*
- Nov.2023, Third Prize in 8th National Cryptography Contest [certificate][list][news] — *Top12.4%, National*
- Sep.2023, Second Prize in Contemporary Undergraduate Mathematical Contest in Modeling [cert.] — *Provincial*
- Sep.2023, Merit-based Scholarships For Outstanding Students — *Top0.15%, School*
- Aug.2023, Third Prize in 16th CISCN Security Project Contest [cert.][report][slide][video] — *Top13.1%, National*
- Jun.2023, Third Prize in 16th CISCN AWDP (Attack with Defense Plus) Contest [certificate] — *Regional*
- Apr.2023, First Prize in 13th MathorCup Mathematical Modeling Challenge [cert.][paper] — *Top5.7%, National*
- Apr.2023, First Prize in 14th Lanqiao Software Development and Algorithm Contest [certificate] — *Provincial*
- Sep.2022, Merit-based Scholarships For Outstanding Students — *Top2.91%, School*
- Jun.2022, Third Prize in 15th CISCN CTF (Capture the Flag) Contest [certificate] — *Regional*
- Jun.2021, Third prize in 12th Lanqiao Software Development and Algorithm Contest [cert.], *Top15.3%, National*
- Sep.2021, Merit-based Scholarships For Outstanding Students — *Top17.61%, School*
- Apr.2021, First Prize in 12th Lanqiao Software Development and Algorithm Contest [certificate] — *Provincial*

Education Degrees

Duke University — **May 2026 (Expected)**

Master of Science Candidate in Medical Physics

Research Focus: Trustworthy AI; Large Language Model; Medical Image Analysis (Base at Duke Kunshan for the first year)

North China University of Technology — **June 2024**

Bachelor of Engineering in Information Security — *GPA: 3.89/4 Grade: 90.88/100 transcript*

Honors: Outstanding Dissertation Award (Rank 1/101, Top 0.99%), Elite Student Scholarship (Rank 1/329, Top 0.30%)

李天昊

tianhao.li | tianhao.li2@duke.edu | 领英 | 谷歌学术 | Orcid | Github

学术界研究经历

威斯康星大学麦迪逊分校 **2024年10月 – 至今**

研究实习 @ 安全基础模型系统实验室（SaFoLab） 美国，威斯康星

- **职责**: 带领科研团队推进AgentFuzzer项目，提出统一的智能体模式描述符AgentSchemaDescriptor，基于此提出首个针对大模型智能体的系统性模糊测试框架，解决了对于智能体安全对齐测试泛化性难以提升的挑战。(将投稿至ICML'25)
- **导师**: Chaowei Xiao教授

浙江大学 **2024年8月 – 2024年10月**

访问研究员 @ 知识引擎实验室 中国，浙江

- **职责**: 带领15人科研团队推进SciSafeEval项目，提出首个面向大模型科学任务安全性的基准评测，在该基准评测下的实验结果指出：包括GPT-4o/o1-preview、LLaMa3.2、QWen2.5等在内的SOTA大语言模型(LLMs)存在生物、化学、医学、物理等科学领域的技术滥用安全风险，同时给出了基于思维链(CoT)技术与小样本学习(Few-shot)技术的有效风险缓解措施。
- **产出**: arXiv:2410.03769 (ICLR'25 Sub) | Huggingface:Tianhao0x01/SciSafeEval | Github:DavidLee528/SciSafeEval
- **导师**: 张强教授，陈华钧教授

高精尖创新研究院 **2022年7月 – 2023年7月**

研究助理 @ 虚拟与增强现实技术创新中心 中国，北京

- **职责**: 带领了一个8人研发团队，开发了基于步态的情报、监视和侦察（ISR）系统。实现了步态识别SaaS原型系统，并提出了包括身份认证、跟踪和近源渗透预警在内的安全特性。负责技术架构设计、技术报告撰写以及研究论文的撰写工作。
- **产出**: 成功申请10万人民币研发经费，发表7篇SCI/EI论文，赢得4项国家级专业比赛，3次口头报告与1篇海报汇报。
- **导师**: 王立军院士

工业界研究经历

天融信 **2024年3月 – 2024年9月**

安全研究员 @ 创新技术研究部 中国，北京

- **职责**: **(i)** 支撑机器学习模型安全评估项目原型系统落地，涉及到对于对抗攻击、投毒攻击、等算法的调试以及原型系统后端评估模块的开发; **(ii)** 支撑网络钓鱼电子邮件检测项目，监督数据集的收集和改进工作。成功应用大型语言模型 (LLM) 和监督微调 (SFT) 开发出强大的检测系统。
- **领导**: 创新研究总经理王玮博士

绿盟科技 **2023年9月 – 2024年3月**

安全研究员 @ 创新研究院天枢实验室 中国，北京

- **职责**: **(i)** 设计绿盟大模型安全评估系统(NSFOCUS LSAS)核心算法与原型系统; **(ii)** 通过进攻性测试，即"红队"测试，提高绿盟风云卫大模型(NSFGPT)和其他私有LLM的安全和隐私性能; **(iii)** 进行文献综述，包括可信人工智能的研究论文和Gartner咨询报告，丰富威胁情报数据库; **(iv)** 撰写LLM安全技术白皮书。
- **影响**: 发布了绿盟大模型安全评估系统(LSAS)，绿盟风云卫大模型NSFGPT以及安全行业大模型SecLLM技术白皮书
- **领导**: 主管安全研究员王星凯博士, 首席创新官&创新研究总经理刘文懋博士

部分文章发表

- Tianhao Li, Jingyu Lu, Chuangxin Chu, Tianyu Zeng, Yujia Zheng, Mei Li, Haotian Huang, Bin Wu, Zuoxian Liu, Kai Ma, Xuejing Yuan, Xingkai Wang, Keyan Ding, Huajun Chen, Qiang Zhang*. SciSafeEval: A Comprehensive Benchmark for Safety Alignment of Large Language Models in Scientific Tasks. *International Conference on Learning Representations, 2025 (ICLR, Submitted)* [PDF]
- Tianhao Li, Weizhi Ma, Yujia Zheng, Xinchao Fan, Guangcan Yang, Ying Li, Lijun Wang, Zhengping Li*. A Survey on Gait Recognition Against Occlusion: Taxonomy, Dataset and Methodology. *PeerJ Computer Science, 2024 (SCI, JCR-Q1, Accepted)* [Poster]
- Weizhi Ma, Ying Li*, Tianhao Li, Haowei Yang, Zhengping Li, Lijun Wang, Junyu Xuan. SFSWTS: EEG Emotion Recognition Based on Spatial-frequency Shifted Windows Time Self-attention Neural Network. *Pattern Recognition, 2024 (SCI, JCR-Q1, Peer Review)*
- Weizhi Ma, Yujia Zheng, Tianhao Li, Zhengping Li, Ying Li, Lijun Wang*. A comprehensive review of deep learning in EEG-based emotion recognition: classifications, trends, and practical implications. *PeerJ Computer Science, 2024 (SCI, JCR-Q1)* [PDF]

- Tianhao Li, Yujia Zheng, Weizhi Ma, Guangshuo Wang, Zhengping Li, and Lijun Wang*. Trustworthy Metaverse: A Comprehensive Investigation into Security Risks and Privacy Issues in Artificial Intelligence-Extended Reality Systems. *SID Symposium Digest of Technical Papers, 2024 (ICDT)*
- Yujia Zheng, Tianhao Li, Weizhi Ma, Jiaxiang Zheng, Zhengping Li*, and Lijun Wang. Unveiling Privacy Challenges: Big Data-Driven Digital Twins in Smart City Applications. *SID Symposium Digest of Technical Papers, 2024 (ICDT)*
- Tianhao Li, Yujia Zheng, Haoan Zhang, Weizhi Ma, and Ying Li*. Research on Real-time Early Warning Mechanism of Aviation Safety Based on Finite State Machine Underlying in QAR Stream Data. *Proceedings of the 2023 7th International Conference on Big Data and Internet of Things, 2023 (BDIoT)* [PDF][Oral][Slide]

开源活动

- NVIDIA/garak(生成式人工智能红队工具，大模型安全领域的Metasploit，2.8K+ Stars)

  (i) 鲁棒性提升: Python解释器版本错误修复#296; Replicate大模型生成器异常修复#401; Detector检测器基类增加异常处理逻辑的增加#566; (ii)新特性引入: 针对LLaVA的多模态越狱攻击#587 (使其成为**世界上首个**具备多模态红队能力的工具); 多模态红队提示词探针架构升级 (#658); 全新探针SciSafeEval的引入 (#948); 生成器网络代理特性 (garak:proxy_dev). (v0.9.0.9, v0.9.0.11, v0.9.0.13)

学术服务

- Reliability Engineering & System Safety, Elsevier *(JCR-Q1, 中科院一区, IF=9.4, Top)* 审稿人
- The 31st International Conference on Computational Linguistics (COLING 2025) 审稿人
- MLCommon/人工智能风险与可靠性(AIRR)工作组 成员/独立贡献者

荣誉奖项

- (2024年08月, 国家级, *Top7.2%*) 第十七届全国大学生信息安全竞赛作品赛二等奖 [证书]
- (2024年07月, 国家级, *Top9.0%*) 星河生成式人工智能安全大赛攻击赛道第8名（共89队）
- (2024年02月, 国家级, *Top?%*) 2024美国大学生数学建模竞赛(MCM)S奖 [论文]
- (2023年11月, 国家级, *Top9.8%*) 第十一届全国大学生数字媒体科技作品及创意竞赛二等奖 [证书]
- (2023年11月, 国家级, *Top12.4%*) 第八届全国密码技术竞赛三等奖 [证书][名单][新闻]
- (2023年08月, 国家级, *Top13.1%*) 第十六届全国大学生信息安全竞赛作品赛三等奖 [证书][报告][幻灯片][演示][新闻]
- (2023年04月, 国家级, *Top5.7%*) 第十三届MathorCup数学建模挑战赛一等奖 [证书][论文]
- (2021年06月, 国家级, *Top15.3%*) 第十二届蓝桥杯全国软件和信息技术专业人才大赛三等奖 [证书]
- (2023年09月, 省部级) 全国大学生数学建模竞赛北京赛区甲组二等奖 [证书]
- (2023年06月, 区域级) 第十六届全国大学生信息安全竞赛实践赛(AWDP)三等奖 [证书]
- (2023年04月, 省部级) 第十四届蓝桥杯全国软件和信息技术专业人才大赛一等奖 [证书]
- (2022年06月, 区域级) 第十五届全国大学生信息安全竞赛实践赛(CTF)三等奖 [证书]
- (2021年04月, 省部级) 第十二届蓝桥杯全国软件和信息技术专业人才大赛一等奖 [证书]
- (2022-2023学年, *Top0.15%*) 荣获优秀学生、学习、创新实践、体育、思想品德奖学金。
- (2021-2022学年, *Top2.91%*) 荣获优秀学生、学习、创新实践、体育、思想品德奖学金。
- (2020-2021学年, *Top17.61%*) 荣获学习、创新实践奖学金。

教育学位

杜克大学 **2026年5月 (预计)**

理学硕士研究生 - 医学物理

研究方向: 可信人工智能；大语言模型；医学图像处理。（首年base昆山杜克）

北方工业大学 **2024年6月**

工学学士 - 信息安全 绩点: *3.89/4* 成绩: *90.88/100* 成绩单

荣誉奖项: 优秀毕业论文 (排名1/101, Top 0.99%), 一等优秀学生奖学金 (排名1/329, Top 0.30%)